

Group 32



USC Viterbi School of Engineering

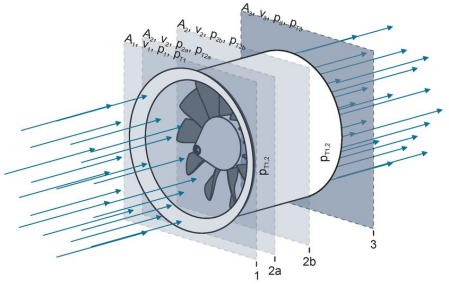
AME 441A

Fall 2025

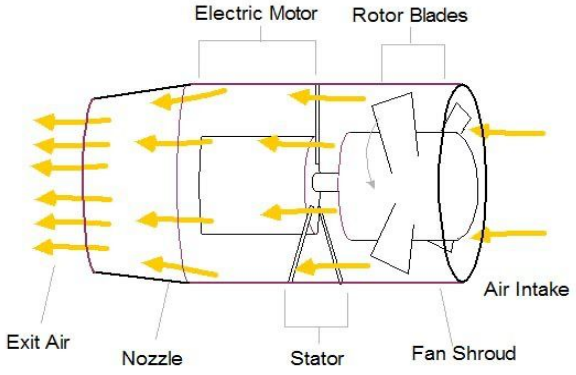
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INTRODUCTION

Problem - Non-ducted propellers are less efficient than ducted counterparts; however, electric ducted propulsors are under-characterized in static conditions, making it hard to compare designs or validate models.



Goal - Build and calibrate a modular static-thrust test stand for open and ducted rotors to generate repeatable thrust-per-watt and non-dimensional performance data. Use this data to prove increased efficiency of ducted vs non-ducted



Approach - Instrumented load path (S-beam), RPM + electrical sensing, LabVIEW control/logging, rapid-swap rotor/duct hardware, safety interlocks.

Impact - Delivers benchmark data for design trade-offs, supports course research, and creates a reusable lab asset for future cohorts.

Theory

We **model static tests** ($V_\infty \approx 0$) with a rigid, steady rotor and flow (tip Mach < ~0.6); the duct acts like an axisymmetric diffuser that raises mass flow and reduces tip vortices. From actuator-disk momentum theory, an ideal disk gives $T = 2 \cdot \rho \cdot A \cdot v_i^2$ and $P_{ideal} = T \cdot v_i$, motivating a static figure of merit $FoM \approx T^{(3/2)} / (P_{in} \cdot \sqrt{(2 \cdot \rho \cdot A)})$.

.

To **compare designs across size/RPM**, we use non-dimensional metrics: $C_T = T/(\rho \cdot n^2 \cdot D^4)$ and $C_{P(elec)} = P_{in}/(\rho \cdot n^3 \cdot D^5)$, where $n = RPM/60$; duct benefit is $Aug = T_{ducted}/T_{open}$ at matched RPM or P_{in}. Scaling terms explain deviations: $V_{tip} = \pi \cdot D \cdot n$, $M_{tip} = V_{tip}/a$, and $Re = \rho \cdot V_{tip} \cdot c/\mu$ (low Re hurts performance; high M_{tip} adds losses/noise).

Data reduction: $T = k \cdot (V_{AI} - V_{zero})$, $P_{in} = V \cdot I$, $\rho \approx P/(R \cdot T)$, then compute C_T, C_P, FoM on steady windows; report honest bars via $\Delta T \approx (V_{range}/2^N)/k$ and standard propagation for C_T (errors from T, n, D, ρ), targeting ~2–5% total in steady runs.

Performance Modeling & Metrics

What we quantify: static thrust, RPM, voltage, current, vibration

Why non-dimensionalize: compare rotors/ducts across sizes and speeds, not just raw thrust

Key outputs: C_T, C_P, Figure of Merit, thrust-per-watt, augmentation vs open prop



Performance Modeling & Metrics — Equation Sheet

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Core definitions

$$n = \frac{\mathrm{RPM}}{60} \quad [\mathrm{rev/s}], \tag{1}$$

$$D: \text{ prop diameter}, \qquad A = \pi\left(\frac{D}{2}\right)^2, \tag{2}$$

$$\rho = \text{air density} \approx \frac{P}{RT} \quad \text{(ideal gas)}, \tag{3}$$

$$P_{\mathrm{in}} = V\,I. \tag{4}$$

Non-dimensional performance

$$C_T = \frac{T}{\rho\, n^2 D^4}, \tag{5}$$

$$C_P^{(\mathrm{elec})} = \frac{P_{\mathrm{in}}}{\rho\, n^3 D^5}, \tag{6}$$

$$\mathrm{FoM}_{\mathrm{static}} \approx \frac{T^{3/2}}{P_{\mathrm{in}}\sqrt{2\,\rho\, A}}. \tag{7}$$

Duct augmentation (static comparison)

$$\mathrm{Aug} = \frac{T_{\mathrm{ducted}}}{T_{\mathrm{open}}} \quad \text{(at matched } P_{\mathrm{in}} \text{ or matched RPM)}. \tag{8}$$

Operating regimes & scaling

$$J = \frac{V_\infty}{nD} \quad \text{(static: } J = 0), \tag{9}$$

$$V_{\mathrm{tip}} = \pi D n, \qquad M_{\mathrm{tip}} = \frac{V_{\mathrm{tip}}}{a}, \tag{10}$$

$$\mathrm{Re} = \frac{\rho\, V_{\mathrm{tip}}\, c}{\mu}. \tag{11}$$

Sensing, calibration, and density correction

$$T = k\,(V_{AI} - V_{\mathrm{zero}}), \tag{12}$$

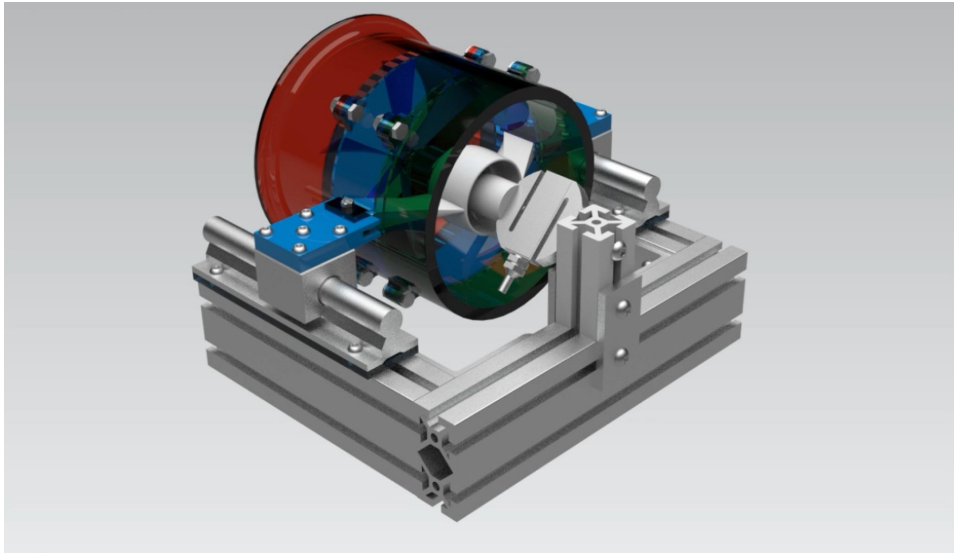
$$\rho(T,P) \approx \frac{P}{RT}, \quad n = \frac{\mathrm{RPM}}{60}, \quad P_{\mathrm{in}} = V\,I. \tag{13}$$

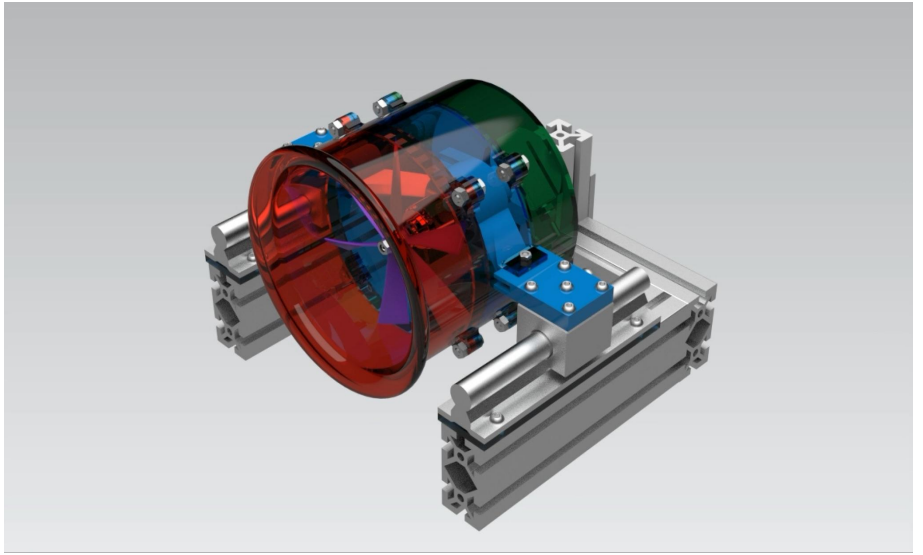
Resolution (quantization-limited thrust example)

$$\Delta T \approx \frac{V_{\mathrm{range}}/2^N}{k}. \tag{14}$$

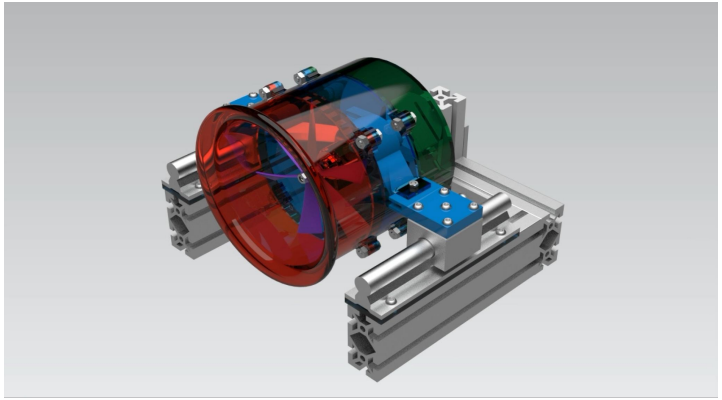
Uncertainty propagation for C_T

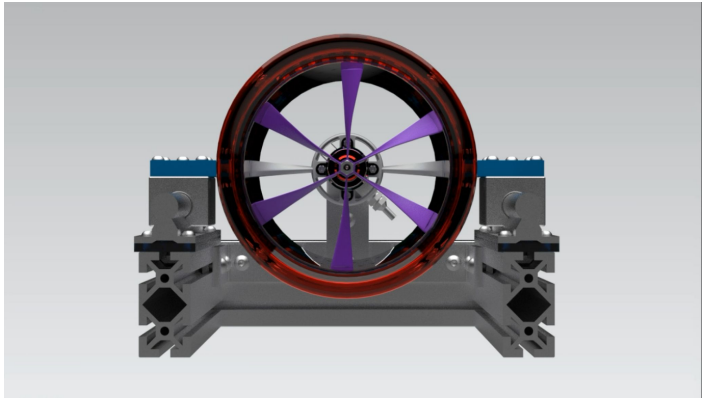
Model



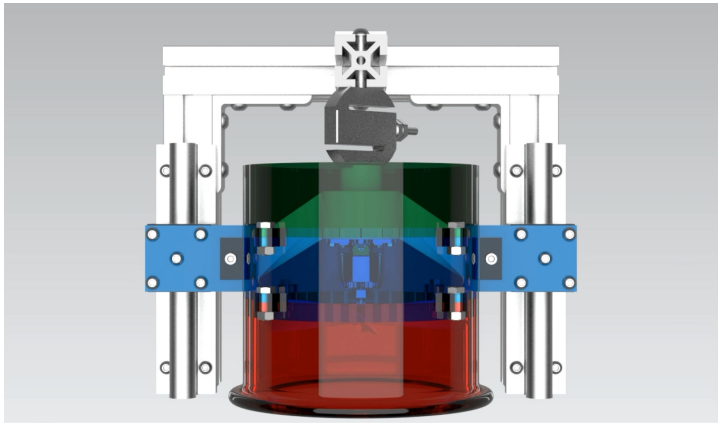


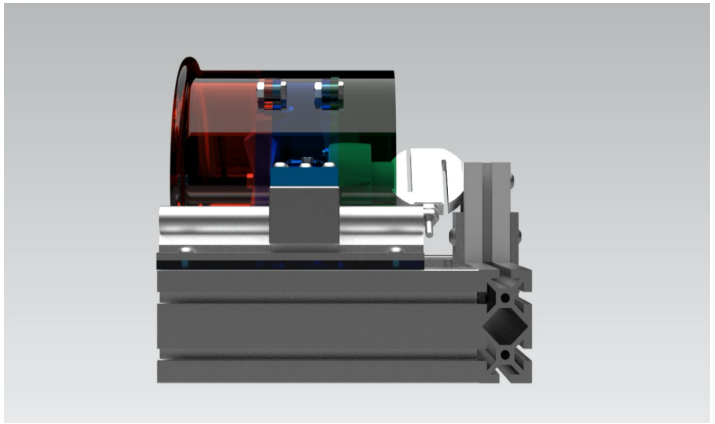
CAD renders of the modular static-thrust rig. An 80/20 aluminum base supports a linear-rail carriage with a blue quick-swap motor/adapter bracket. The prop sits inside a segmented duct (red intake lip, blue center ring, green diffuser) bolted to the mount, with the load path aligned to an S-beam/load-cell bracket for thrust measurement. The left view shows the full assembly and load path; the right view highlights the duct geometry and quick-swap interface for rapid configuration changes.

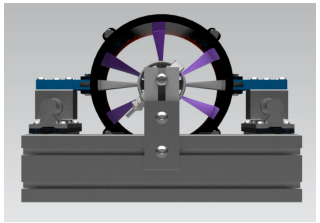




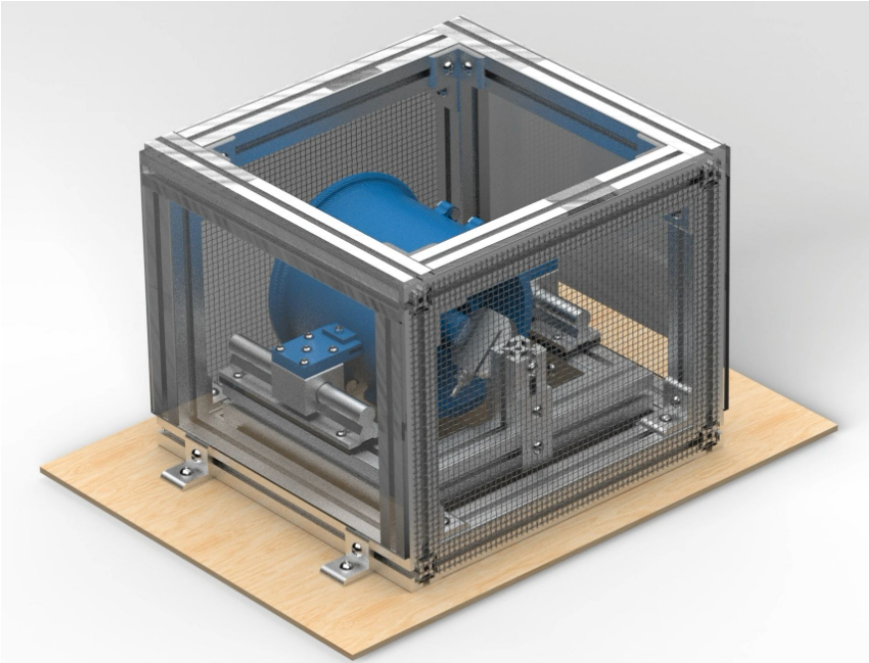
Model







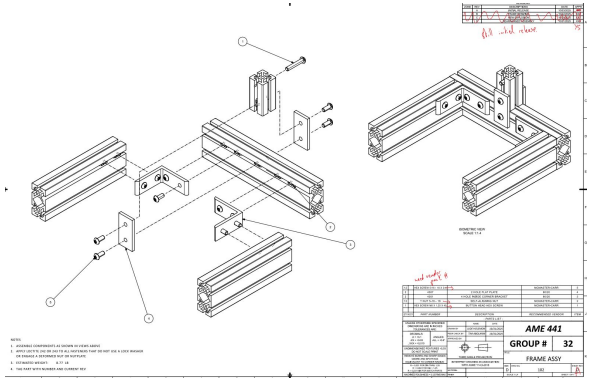
Model

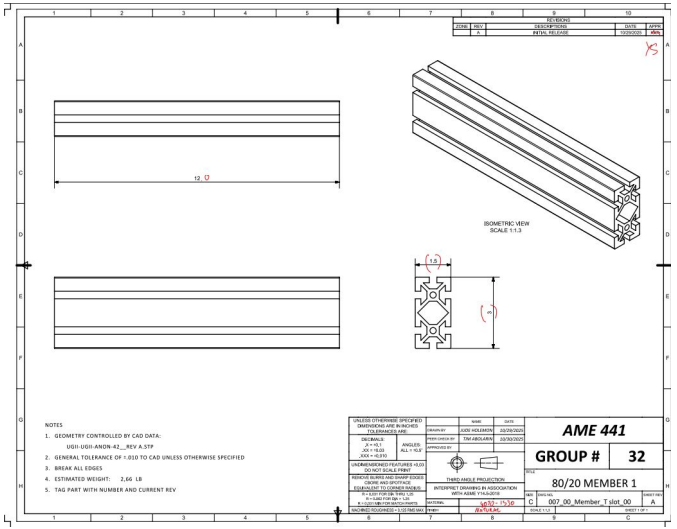


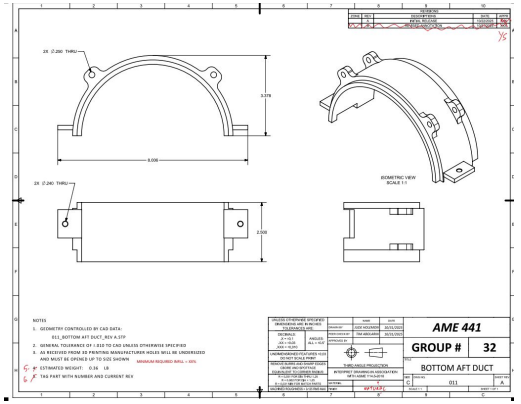
GOON Cage (Guarded Overhead & Outer Netting)

The test stand is enclosed in a rigid 80/20 frame with full-perimeter steel mesh panels and a solid overhead guard, anchored to a plywood base for energy containment and stability. The GOON cage maintains clear sightlines and airflow while preventing blade fragments, loose hardware, or tool drops from exiting/entering the test volume. A front service panel (interlocked in final build) and external E-stop keep operators outside the hazard zone during arm, run, and shutdown.

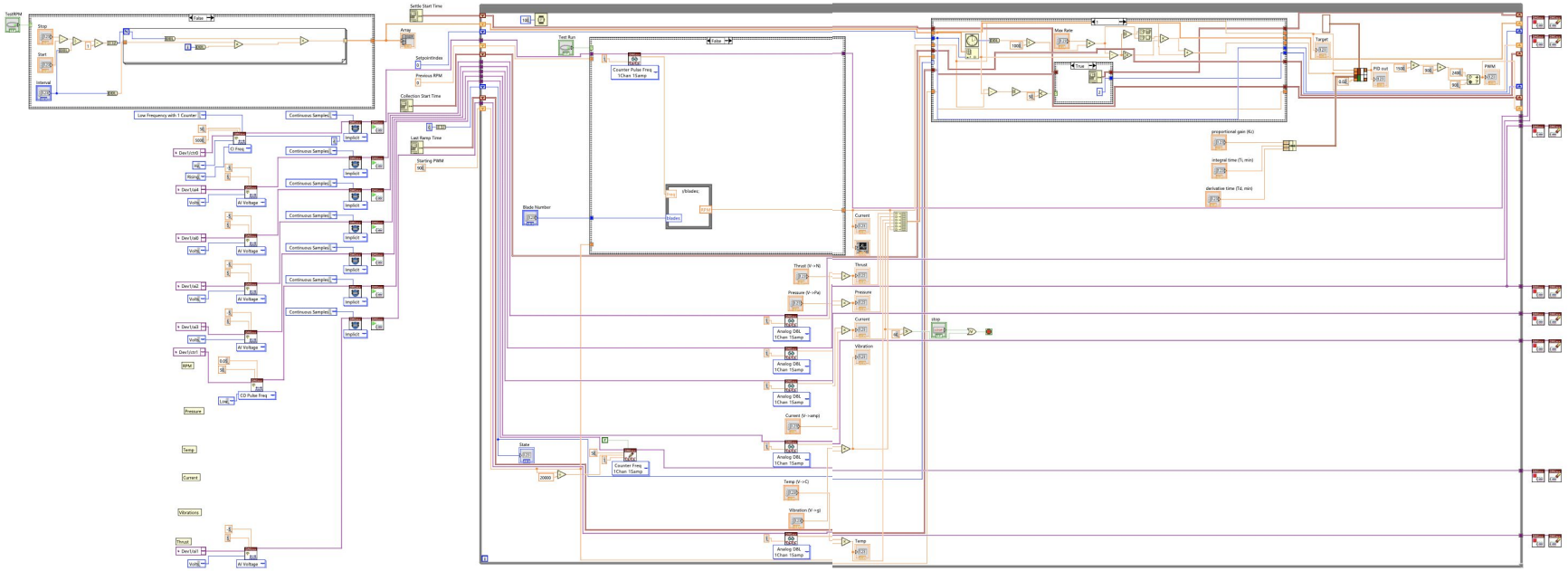
Setup / Drawings







Data Flow : LabView



Data Flow : LabView

User Input

RPM Inputs

- Max RPM
- Min RPM
- Interval

→ RPM Array

- Max Rate of RPM Change
- Blade Number

PID Inputs

- Kc
- Td
- Ti

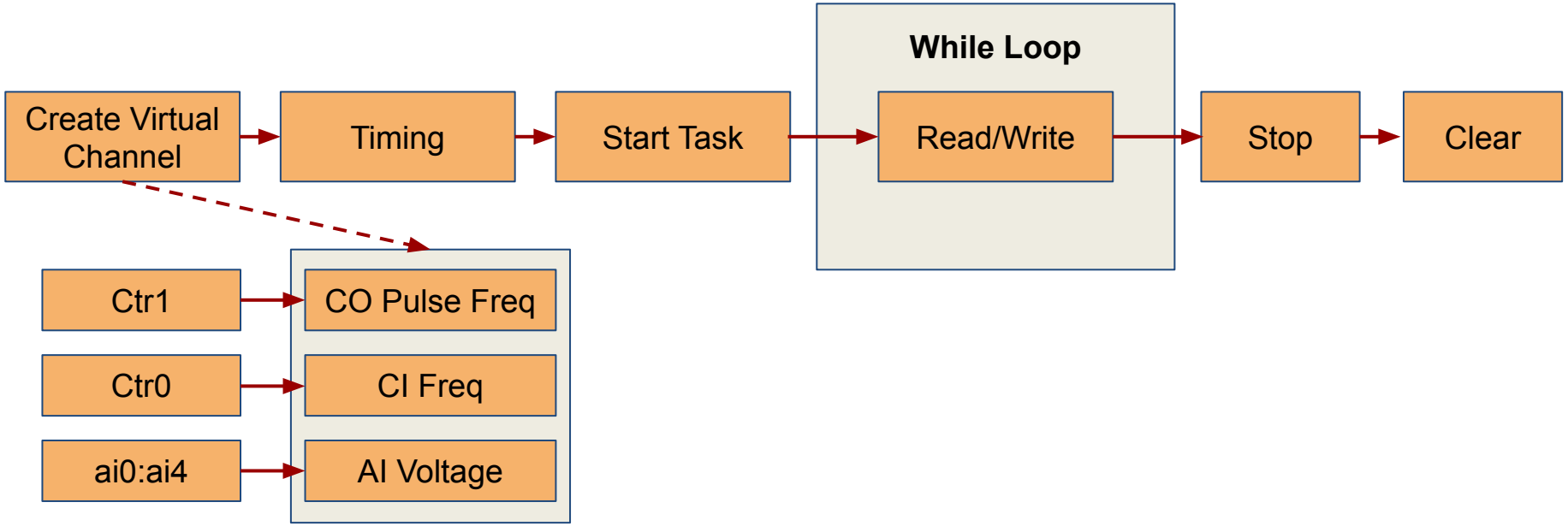
Timing

- Settle
- Collection

Volt to Engineering Units Conversion Factor for Analog Inputs

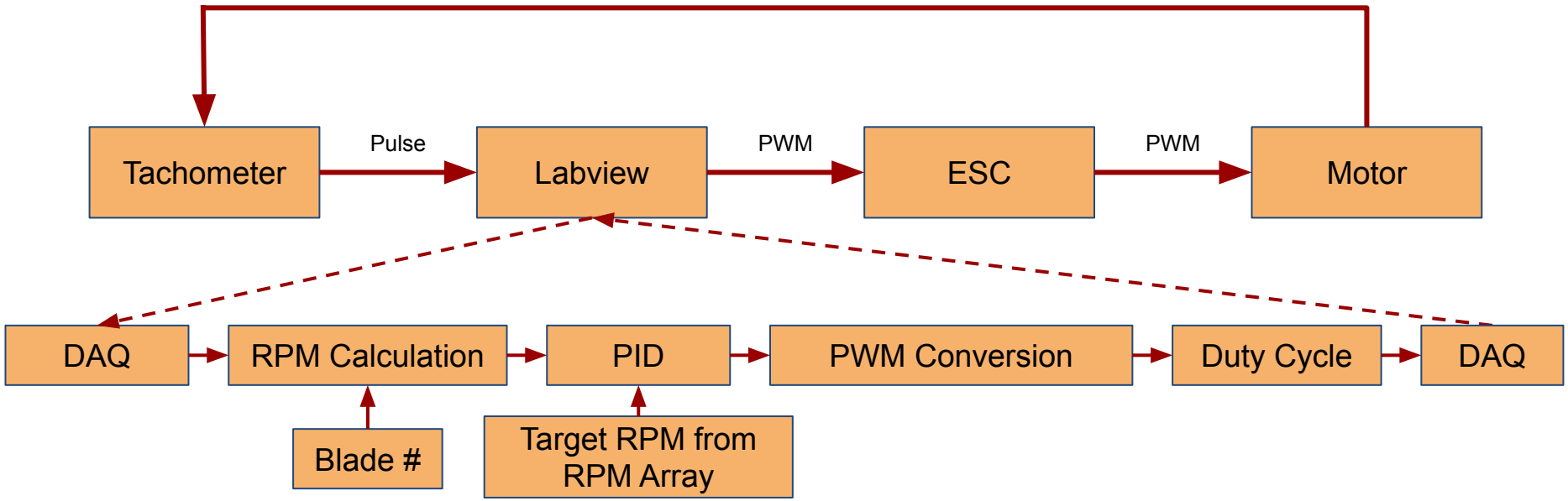
- Thrust
- Temp
- Pressure
- Vibration
- Current

Data Flow : LabView



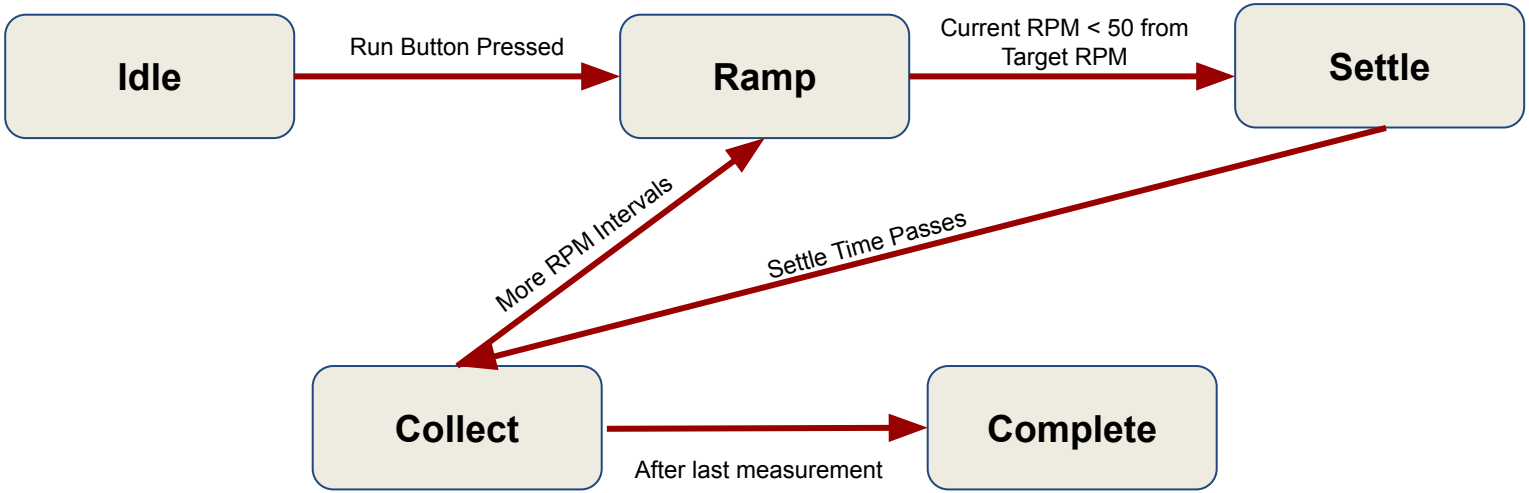
Data Flow : LabView

RPM Control System

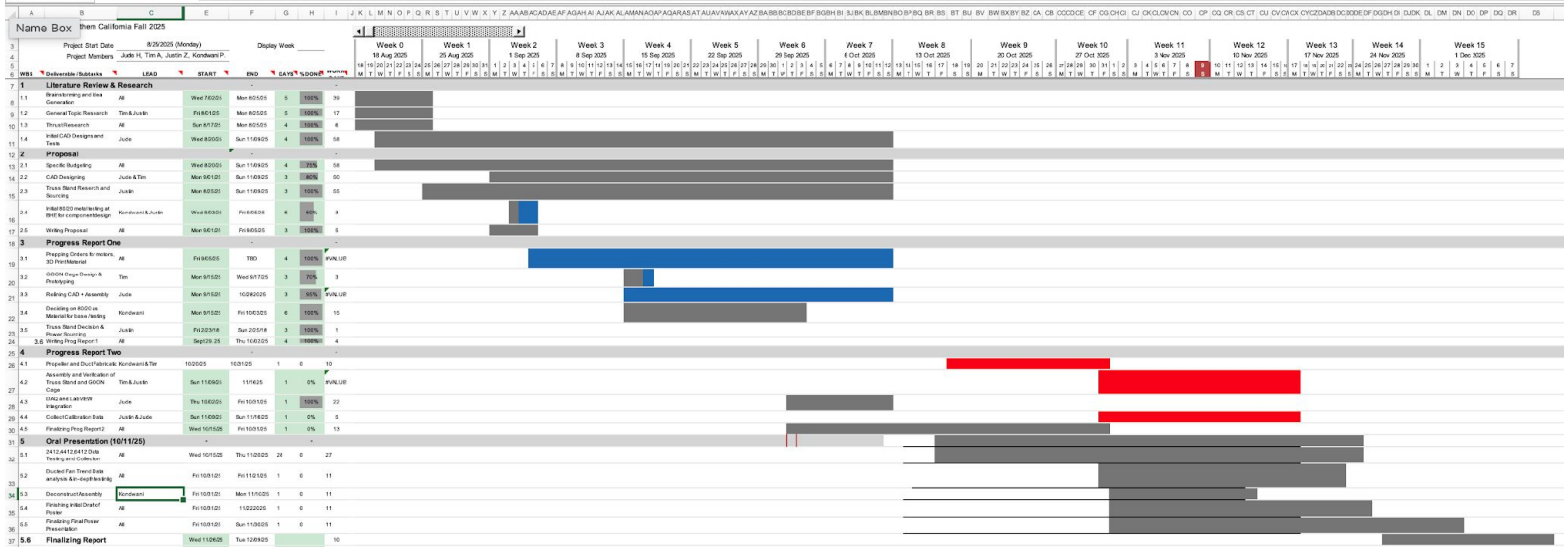


Data Flow : LabView

Five Cases to Automate

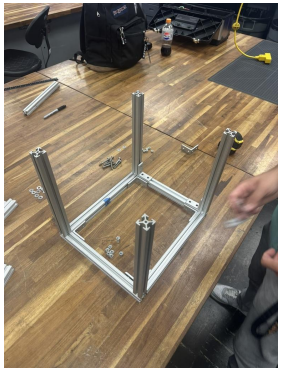


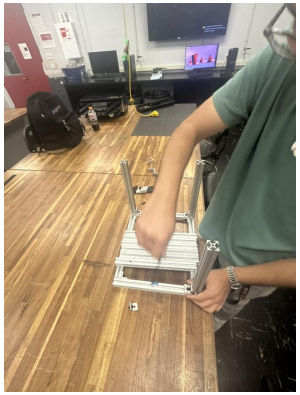
Process



Behind earlier proposed schedule

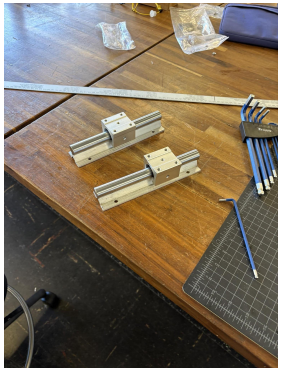
Process

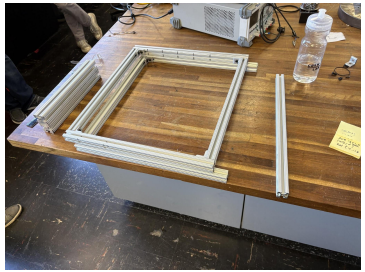




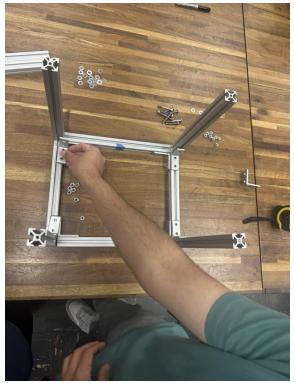


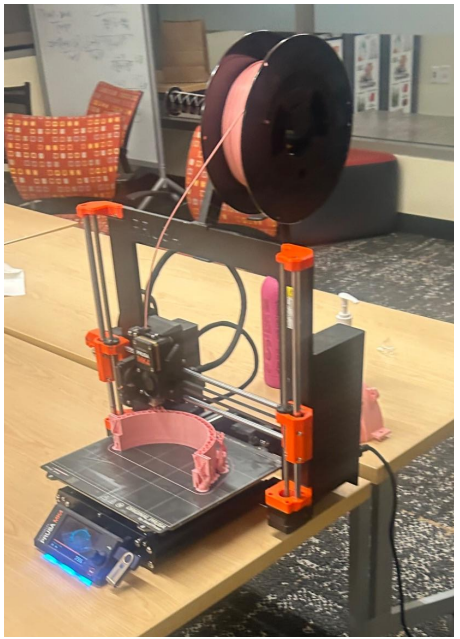












Thank You

